

```
saketh@DESKTOP-0Q0P5LQ:~/OSLAB$ nano m23.c
saketh@DESKTOP-0Q0P5LQ:~/OSLAB$ gcc m23.c -o m23
saketh@DESKTOP-0Q0P5LQ:~/OSLAB$ ./m23
```

```
=====
COMMAND HISTORY BUFFER
=====
```

Commands:

add <command>	- Store command
history	- Display history
get <number>	- Retrieve command
clear	- Clear history
exit	- Exit

```
history-shell> hi
Invalid command.
```

```
history-shell> add
Invalid command.
```

```
history-shell> clear
History cleared.
```

```
history-shell> |
```

```
saketh@DESKTOP-OQ0P5LQ:~$ cd OSLAB
saketh@DESKTOP-OQ0P5LQ:~/OSLAB$ nano m11.sh
saketh@DESKTOP-OQ0P5LQ:~/OSLAB$ chmod +x m11.sh
saketh@DESKTOP-OQ0P5LQ:~/OSLAB$ ./m11.sh
```

```
=====
SINGLE QUOTES AND DOUBLE QUOTES
=====
```

```
----- SINGLE QUOTES -----
```

```
Hello World
Hello $NAME
The value of $NAME is not expanded.
Stored single-quoted string:
This is a $NAME literal string.
Literal content:
Hello      World      Linux
```

```
----- DOUBLE QUOTES -----
```

```
Hello World
Hello Shanmukha
The value of NAME is: Shanmukha
Stored double-quoted string:
Hello Shanmukha
String with spaces:
Hello      Linux      World
```

```
----- VARIABLE EXPANSION -----
```

```
Single quotes: $NAME
Double quotes: Shanmukha
```

```
----- NESTED TOKENS -----
```

```
Outer quotes with variable: Linux Shell
Command result: Nested Command
```

```
----- QUOTED COMMANDS -----
```

```
Command stored as:
echo Hello Linux
```

```
Executing command:
Hello Linux
```

```
----- EDGE CASES -----
```

```
Length of single-quoted empty string: 0
Length of double-quoted empty string: 0
Special characters: $ * ? ! # @
Special characters: $ * ? ! # @
```

```
All quote tests completed successfully.
saketh@DESKTOP-0Q0P5LQ:~/OSLAB$ nano m12.sh
saketh@DESKTOP-0Q0P5LQ:~/OSLAB$ bash m12.sh
=====
          ESCAPE SEQUENCE DEMONSTRATION
=====

----- ESCAPED SPACES -----
Hello Linux
Hello Linux World

----- ESCAPING SPECIAL SYMBOLS -----
Dollar symbol: $
Asterisk: \*
Question mark: \?
Exclamation mark: \!
Hash symbol: \#
Backslash: \

----- ESCAPED CHARACTERS -----
Line1
Line2
Column1 Column2
FirsSecond
World

----- PRESERVING CHARACTERS -----
Hello $USER \* \? \#

----- ESCAPED FILE NAME -----
Created file:
-rw-r--r-- 1 siddharth siddharthreddy 0 Oct  1 18:12 'My Test File.txt'

----- COMPLEX INPUT -----
Hello\ World\ $USER\ \*\ \?

----- PARSER OUTPUT -----
Normal:
Hello World
Escaped:
Hello World
Quoted:
Hello World

Escape sequence testing completed.
```